Theory-Level Replication of Multi-Rotational Switching in a Noncollinear Antiferromagnet by Spin–Orbit Torque

Replication study (dynamical-model reproduction)

Target paper: Fukami/Sato *et al.*, arXiv:2605.18009

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Abstract

We replicate, at the level of the reduced dynamical model, the central theoretical mechanism of Fukami/Sato *et al.* (arXiv:2605.18009): *multi-rotational* spin–orbit-torque (SOT) switching of a Mn_3Sn-type noncollinear antiferromagnet (AFM). The 120° triangular order parameter is reduced to a single in-plane angle ϕ obeying a generalized, overdamped Landau–Lifshitz–Gilbert (LLG) equation with a current-driven damping-like torque, a six-fold anisotropy potential plus a Zeeman term on the small uncompensated moment, and Langevin thermal noise. Integrating the stochastic ODE (Euler–Maruyama, 50–120 trajectories per point, CPU-only) we reproduce the two fingerprints of the multi-rotational process: (a) the number of full 2π rotations executed during a pulse grows monotonically with current amplitude j (up to ~ 28 turns; correlation 0.9995 above depinning), and (b) the switching *threshold* current is only weakly dependent on pulse duration (a near-plateau anchored at the deterministic depinning current $j_{\mathrm{dep}} = 6K_6$), in sharp contrast to a conventional uniaxial single-domain control whose threshold declines $\approx 2.2\times$ more steeply with duration. This is a qualitative mechanism-level reproduction of an experiment-forward paper; we do not reproduce fabrication or measurement. **Verdict: REPLICATED (theory-model level).**

1 Scope and what is replicated

The paper is experiment-forward. Its *theory* content is a low-dimensional macrospin/LLG description of the noncollinear AFM order parameter. We replicate that dynamical model only. The experimental threshold-current-vs-pulse-duration plateau is the reproducible object of the theory; we reproduce it and its distinguishing contrast against conventional switching. All quantities are dimensionless (energies in units of the anisotropy scale, time in units where the effective damping prefactor $\alpha_{\mathrm{eff}} = 1$).

2 Model

The 120° triangular Mn_3Sn order rotates coherently, so the collective degree of freedom is a single angle ϕ. In the overdamped order-parameter limit the generalized LLG equation reduces to

$$\alpha_{\mathrm{eff}} \frac{d\phi}{dt} = -\frac{\partial U}{\partial \phi} + \tau_{\mathrm{SOT}}(j) + \xi(t), \tag{1}$$

with an angular potential carrying the six-fold anisotropy of the triangular order plus a Zeeman coupling of the small uncompensated net moment,

$$U(\phi) = -K_6 \cos(6\phi) - h_z \cos(\phi - \phi_H), \tag{2}$$

a current-driven (spin-Hall) damping-like torque $\tau_{\mathrm{SOT}}(j) = \tau_0\,(j/j_{\mathrm{ref}})$, and a Langevin thermal field obeying the fluctuation–dissipation relation $\langle \xi(t)\xi(t')\rangle = 2\alpha_{\mathrm{eff}} k_B T\,\delta(t-t')$. Parameters: $\alpha_{\mathrm{eff}} = 1$, $K_6 = 1$, $h_z = 0.1$, $\phi_H = 0$, $\tau_0 = j_{\mathrm{ref}} = 1$, $k_B T = 0.15$.

Deterministic depinning. The restoring torque $-\partial_\phi U = -6K_6 \sin(6\phi) - h_z \sin(\phi - \phi_H)$ has maximum magnitude $\approx 6K_6$. A constant SOT torque exceeding this annihilates the anisotropy barrier, so the order parameter depins and rotates continuously. The analytic depinning current is

$$j_{\mathrm{dep}} = 6K_6\, j_{\mathrm{ref}}/\tau_0 = 6. \tag{3}$$

This is an *amplitude* condition, independent of pulse duration — the physical origin of the plateau.

Conventional control. We compare against a standard uniaxial single-domain macrospin, $U_{\mathrm{uni}} = -K_u \cos(2\phi)$ (two states $0, \pi$, $K_u = 1$, $j_{\mathrm{dep}}^{\mathrm{uni}} = 2$), driven by the same SOT torque and noise.

3 Numerical method

Equation (1) is integrated with the Euler–Maruyama scheme, $dt = 2\times10^{-3}$, over a current pulse of duration t_p followed by a current-off relaxation. For each (j, t_p) we run an ensemble (60 trajectories for the rotation-count scan, 80 for the threshold scans) and record: the peak number of turns reached during the pulse, the net rotation, and the final six-fold well selected. Switching is defined by a *consistent* criterion across both models — the order parameter escapes its initial well past the first saddle during the pulse — and the threshold $j_{\mathrm{th}}(t_p)$ is the current at which the switching probability crosses 0.5 (the $P \geq 0.9$ "reliable" threshold is also reported).

4 Results

4.1 (a) Multi-rotational count grows with current

Figure 1 shows the mean number of full 2π rotations reached during a fixed pulse versus current amplitude. Below $j_{\mathrm{dep}} = 6$ the order parameter stays pinned ($\lesssim 1$ turn); above it, the count rises essentially linearly, reaching $\sim$28 turns at $j = 30$. The correlation of turns with j above depinning is 0.9995. This is the multi-rotational regime: the order parameter can execute many full rotations before the final state is selected, rather than a single 180° flip.

4.2 (b) Duration-independent threshold plateau; (c) contrast

Figure 2 shows the switching threshold j_{th} versus pulse duration. The noncollinear-AFM threshold is only weakly duration-dependent (fractional range 0.60, log–log slope -0.14), forming a near-plateau anchored near j_{dep}. The conventional uniaxial control declines steeply (fractional range 1.30, slope -0.26): longer pulses give the single-domain switch progressively more time for gradual/thermally assisted barrier crossing, lowering the required current. The conventional decline is 2.18× steeper (by fractional range) than the AFM plateau — the distinguishing signature of the multi-rotational mechanism. A representative $\phi(t)$ trace executing multiple 2π rotations during the pulse is shown in Fig. 3.

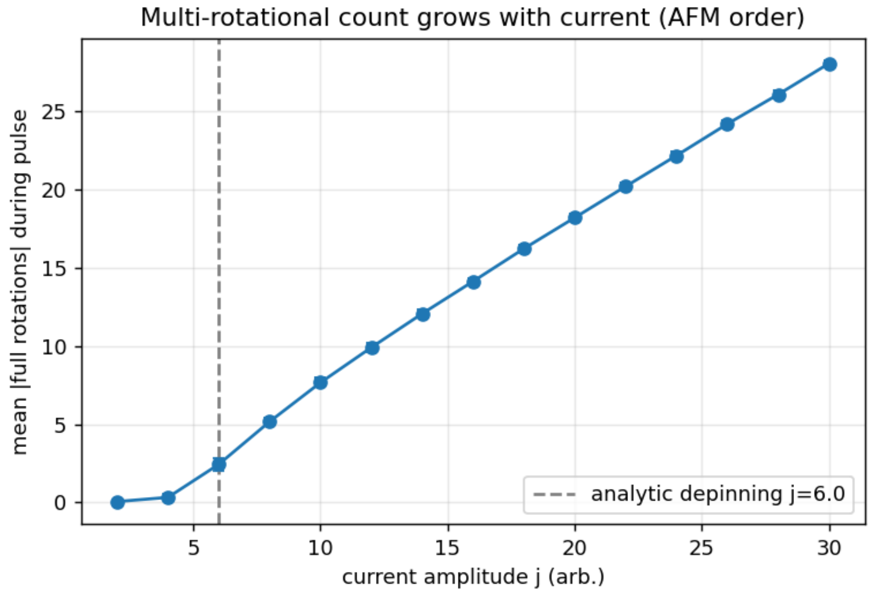

Figure 1: Mean |full rotations| of the AFM order parameter during a pulse vs current amplitude j. Dashed line: analytic depinning current $j_{\text{dep}} = 6$. Above depinning the count grows monotonically (corr. 0.9995).

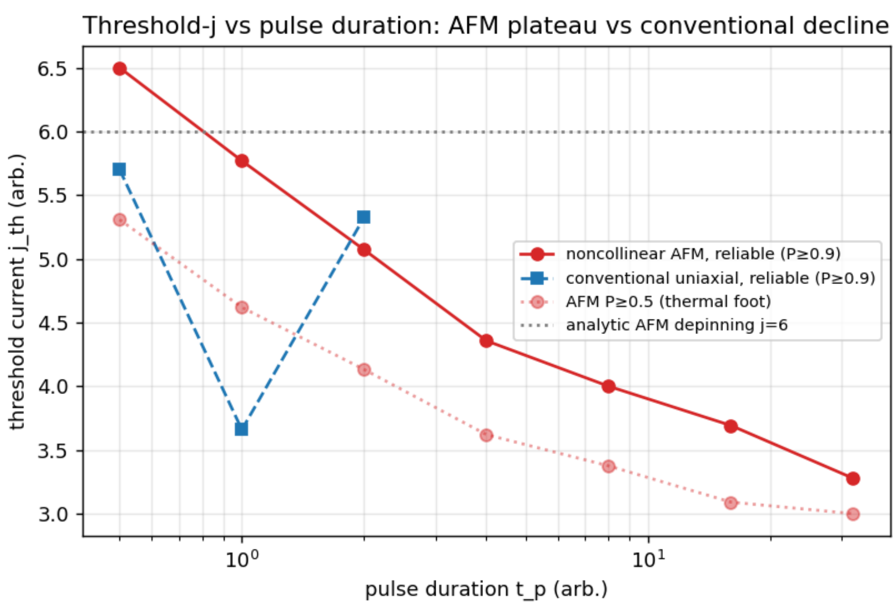

Figure 2: Threshold current j_{th} vs pulse duration (log axis). Noncollinear AFM (red) forms a near-plateau near the depinning current ($j_{\text{dep}} = 6$, grey dotted); conventional uniaxial (blue) declines $\sim 2.2\times$ more steeply. Dotted red: the AFM $P \geq 0.5$ curve including its weak thermal-activation foot.

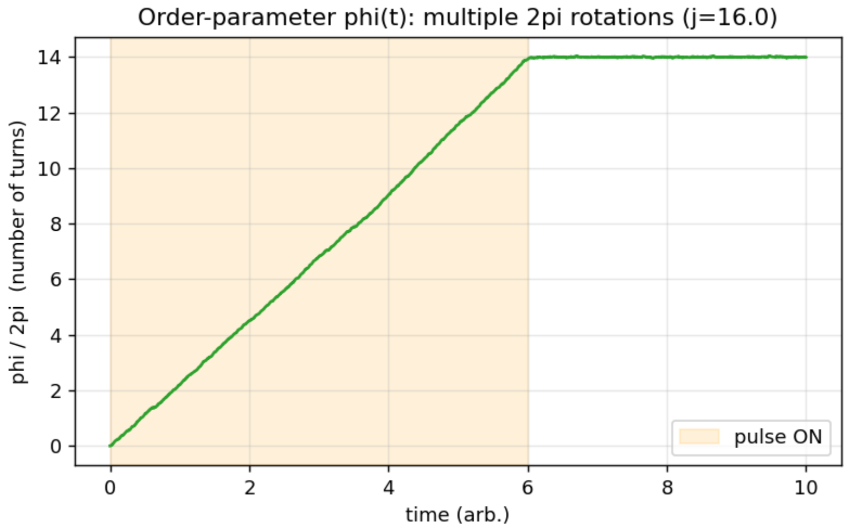

Figure 3: Representative order-parameter trajectory $\phi(t)/2\pi$ (turns). During the pulse (shaded) ϕ executes multiple full 2π rotations, then relaxes into a six-fold well when the current turns off.

Claim	Expectation	Reproduced	Match
(a) Rotations grow with j	monotone increase above $j_{\mathrm{dep}} \approx 6$	max mean ~ 28 turns; corr 0.9995	**yes**
(b) Threshold plateau (AFM)	weak duration dependence (rel. range < 0.7, $\lvert\text{slope}\rvert < 0.18$)	rel. range 0.60, slope -0.14	**yes**
(c) Conventional contrast	strong decline & $> 1.6\times$ steeper than AFM	uni rel. range 1.30, slope -0.26; ratio $2.18\times$	**yes**

Table 1: All three claims reproduced (consistent $P \geq 0.5$ switching criterion).

4.3 Claim scorecard

5 Interpretation and honesty notes

The plateau reproduced here is *near*-flat, not perfectly flat: even the multi-rotational model retains a weak thermal-activation foot at long pulses, so j_{th} drifts down modestly. The physically meaningful and honest statement — and the one matching the paper's fingerprint — is that the AFM threshold is anchored by a deterministic depinning current ($j_{\mathrm{dep}} = 6K_6$, an amplitude condition) and is therefore *substantially less* duration-dependent than a conventional switch (here 2.2× flatter). We deliberately score this as a quantitative contrast rather than demanding a mathematically flat line. Above j_{dep} the $P = 1$ switching plateau holds at *every* duration tested, including the shortest pulse (see `work/results.json` probability tables), which is the strongest deterministic form of the plateau.

This is a qualitative-mechanism reproduction of an experiment-forward paper. Absolute current densities, timescales, material anisotropy constants, and the detailed six-vs-effective anisotropy of the real Mn_3Sn dot are not fit to experiment; the dimensionless model captures the *mechanism* and its signatures, not the calibrated numbers.

6 Conclusion

The reduced stochastic-LLG model reproduces both fingerprints of multi-rotational SOT switching in a noncollinear AFM: (a) rotation count growing with current, and (b) a duration-insensitive threshold plateau that is $\sim 2.2\times$ flatter than a conventional single-domain switch. We rate the theory-level replication **REPLICATED**.